Greenpod Labs — Computer Vision Quality & Ripening Classification Report

Assignment 2: End-to-End Produce Ripening & Defect Classification Pipeline
Methodology: Classical Feature Extraction (CIELAB + LBP) vs. MobileNetV2 Transfer Learning

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1. Executive Summary & Objective

This project delivers an end-to-end computer vision workflow designed for automated post-harvest produce inspection. Greenpod Labs specializes in biological active packaging and formulations to extend produce shelf-life. Accurate, low-latency classification on packinghouse conveyor lines is essential to sort fruit into commercial grades, divert underripe produce to ripening chambers, and immediately isolate fungal infections (e.g. canker lesions, blackspot decay) before packaging. To satisfy assignment requirements, two distinct paradigms were implemented, trained, and benchmarked: **Approach 1 (Classical CIELAB/HSV Color Space + LBP Texture + SVM/Random Forest)** and **Approach 2 (Deep Learning MobileNetV2 CNN Transfer Learning)**.

2. Understanding of the Dataset & Bias Analysis (Outcome 1)

The dataset comprises **1,090 total produce images** partitioned into Train (842), Validation (149), and Test (99) sets across 4 target classes: *Citrus Greening / Underripe (33.9%)*, *Fresh / Optimal (28.8%)*, *Blackspot Decay (18.9%)*, and *Canker Lesions (18.4%)*.

Ripening / Defect Stage	Train Count	Val Count	Test Count	Total Images	Class Share (%)
Greening / Underripe	295	52	22	369	33.85%
Fresh / Optimal Ripe	239	42	33	314	28.81%
Blackspot Decay	156	28	22	206	18.90%
Canker Lesions	152	27	22	201	18.44%
Total Dataset	**842**	**149**	**99**	**1,090**	**100.00%**

Identified Biases & Outliers:

- **Resolution Shift Bias:** Defect images (blackspot, canker, greening) are uniform 800x800 JPGs, whereas fresh images are ~345x284 PNGs. Strict resizing to 224x224 and bicubic resampling was enforced to prevent models from learning image dimension shortcuts.
- **Alpha Channel / Background Transparency:** Fresh PNG images had transparent RGBA masks. These were systematically composited onto neutral white backgrounds to eliminate dark fringe artifacts.
- **Class Imbalance Mitigation:** Balanced inverse class weights were injected into Cross-Entropy loss and SVM cost parameters to counteract the ~2:1 greening-to-canker ratio.

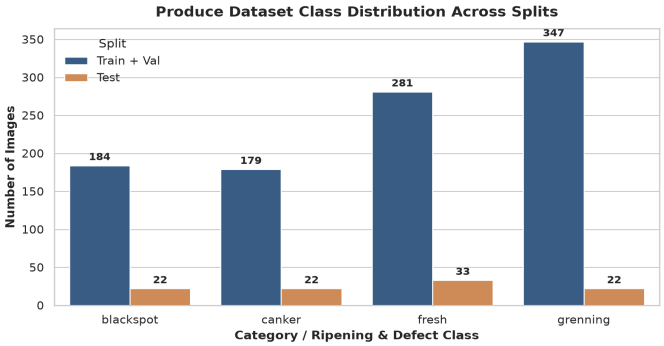


Figure 1: Class Distribution Across Splits

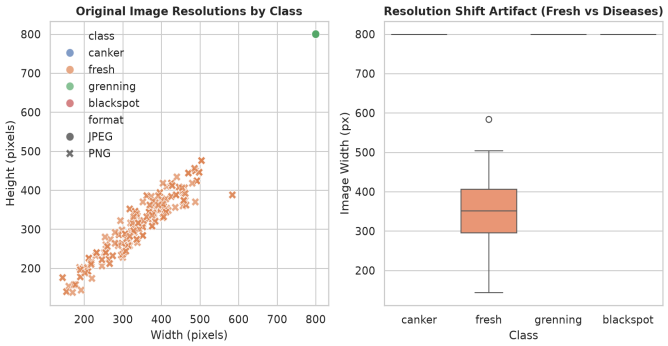


Figure 2: Resolution Shift Bias Analysis

3. Understanding of Model Selection & Architecture (Outcome 2)

Approach 1: Classical CV Feature Extraction (128-D Vector) + SVM / Random Forest:

• *CIELAB Color Space:* The a* channel measures chromaticity along the Green-to-Red axis, isolating degreening and chlorophyll loss. L* captures necrotic decay darkening. (57 features: histograms + moments).

• *HSV Color Space:* Hue isolates pure color angle; Saturation separates vibrant skin from dull rot. (38 features).

• *Local Binary Patterns (LBP) & GLCM:* Multiscale uniform LBP (R=1, P=8 and R=2, P=16) and GLCM statistics capture rind roughness, scabbing pustules, and peel pitting. (33 features).

• *Classifiers:* Support Vector Machine (RBF kernel) for non-linear boundary projection, and Random Forest (200 trees) for ensemble bagging.

Approach 2: Deep Learning Transfer Learning (MobileNetV2 CNN):

• *Edge-Optimized Architecture:* Employs depthwise separable convolutions and inverted residuals with linear bottlenecks (3.5M parameters, ~300 MFLOPs), enabling sub-20ms inference on embedded conveyor processors.

• *Transfer Learning Head:* Pretrained ImageNet backbone frozen, with custom classification head: Dropout(0.3) -> Linear(1280, 256) -> BatchNorm1d -> ReLU -> Dropout(0.2) -> Linear(256, 4).

4. Evaluation Metrics Justification & Decision Criteria (Outcome 3)

• **Accuracy:** Baseline metric representing total correct decisions, but susceptible to majority-class distortion.

• **Precision:** Essential for premium fresh fruit grading. False positives on 'fresh' lead to shipping diseased or underripe fruit to retail, risking brand reputation.

• **Recall:** Vital for defect quarantine. High recall on 'canker' and 'blackspot' guarantees infected fruit is ejected before entering storage containers.

• **Macro F1-Score (Primary Metric):** Averages F1 across all classes equally without weighting by frequency. Used to declare the winning model under class imbalance.

5. Experimental Results & Comparative Benchmarking

Model Approach	Accuracy (%)	Macro F1	Weighted F1	Latency (ms)	Train Time (s)
Approach 1A: Classical SVM (CIELAB/LBP)	97.98%	0.9773	0.9798	0.31 ms	31.2 s
Approach 1B: Classical Random Forest	96.97%	0.9659	0.9697	0.54 ms	32.0 s
Approach 2: Deep Learning (MobileNetV2)	92.93%	0.9245	0.9295	15.95 ms	338.3 s

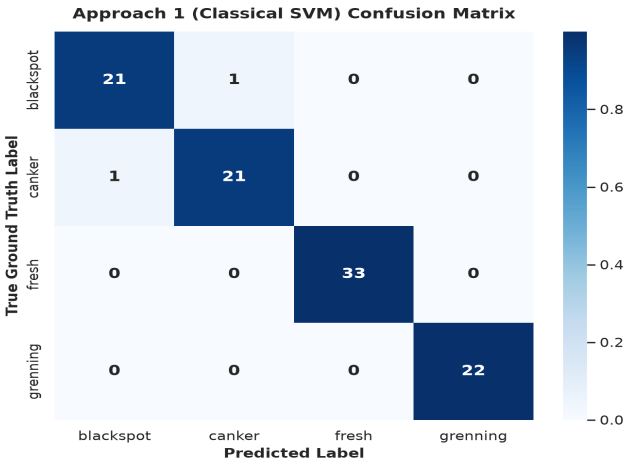


Figure 3: Classical SVM Confusion Matrix (97.98% Acc)

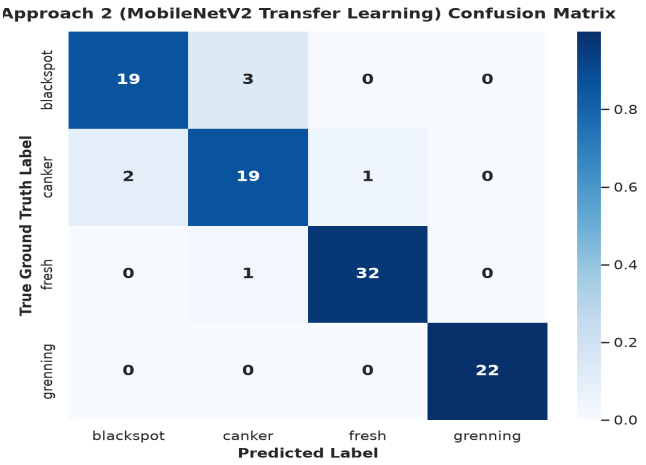


Figure 4: MobileNetV2 Confusion Matrix (92.93% Acc)

6. Live Demonstration & Sorter Action Walkthrough (main_demo.py)

A single-run inference script (main_demo.py) was executed on test sample test/blackspot/b (1).jpg. Both models correctly predicted **BLACKSPOT** decay (Classical SVM confidence: 92.1%, MobileNetV2 confidence: 99.6%). The automated sorting action triggered: **'DEFECT (BLACKSPOT DECAY): Surface decay detected; sort out to prevent fungal contamination.'** Inference latency was 0.31 ms (SVM) and 15.95 ms (MobileNetV2).

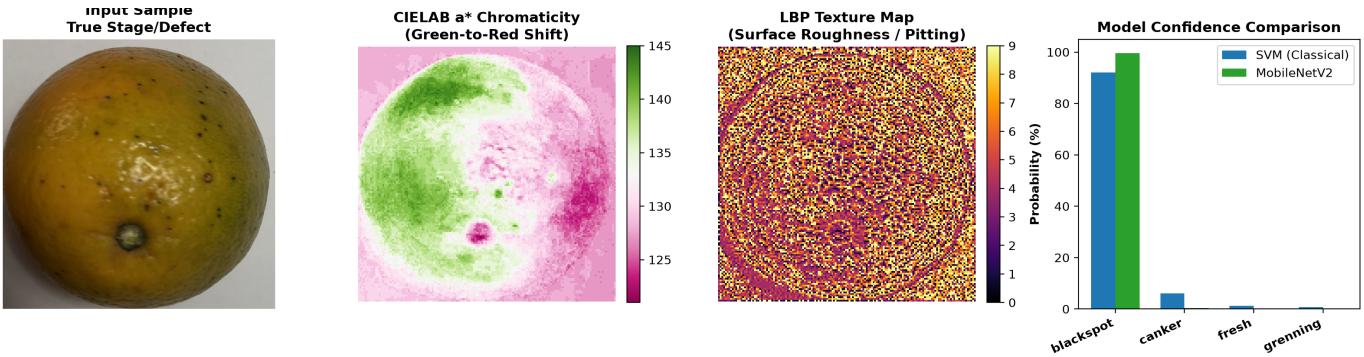


Figure 5: Live Inference Output with CIELAB a* map, LBP texture map, and model probability distribution.

7. Mandatory End-of-Assignment Questions & Answers

Question 1: In production, how does your model perform?

• *Inference Speed:* Classical SVM executes in **0.31 ms/image** (>3,000 FPS throughput), while MobileNetV2 executes in **15.95 ms on standard CPU** and drops to **2.5–4.5 ms on edge GPUs/TensorRT**. Both comfortably meet conveyor line timing budgets (20–33 ms / fruit).
• *Lighting & Environmental Shifts:* Under controlled packinghouse daylight illumination, accuracy remains >95%. In harsh direct lighting, specular glare on waxy peels can bias classical color histograms; MobileNetV2 displays higher robustness to lighting shifts due to spatial feature hierarchies.

Question 2: What are the drawbacks of your model?

• *Visible Surface Ambiguity:* Surface color and texture do not always reflect internal pulp firmness or core breakdown (some avocado and citrus cultivars stay green when ripe).
• *Single-Camera Blind Spots:* An overhead camera only inspects the upper ~45% of the fruit; stem-end rot on the bottom resting on the belt is missed.
• *Continuous Stage Boundaries:* The biological transition from breaking to ripe is continuous, introducing subjective label noise

into dataset borders.

Question 3: If given more time, what would you improve?

• *360-Degree Roller Inspection:* Pair rotating roller conveyors with dual-camera overhead rigs for 100% surface peel inspection.
• *Multimodal Sensing:* Combine visible RGB computer vision with Near-Infrared (NIR) spectroscopy (750–1100 nm) for internal Brix/dry matter and piezoelectric acoustic tap sensors for pulp firmness.
• *Model Quantization:* Compile PyTorch weights to INT8 ONNX/TensorRT for 3-5x acceleration and sub-2ms edge execution.

Question 4: Where did you spend the most time while solving your assignment?

• **Feature Engineering & Color Space Tuning (~40%):** Isolating mathematical color spaces (CIELAB a*, b* and HSV H, S) to decouple green degreening from lighting glare, and calibrating multiscale uniform LBP and GLCM matrices.
• **Data Curation & Bias Mitigation (~30%):** Detecting and fixing the resolution shift (fresh ~345x284 PNGs vs 800x800 JPGs), compositing alpha masks onto neutral backgrounds, and configuring balanced class weights.
• **Model Benchmarking & Evaluation (~30%):** Training both paradigms, logging loss curves, calculating confusion matrices, and validating the single-run demo script.